



Home Work

DevOps

HW_L3_04_Ansibl

e

۱. فایل تمرین را در پنل خود آپلود کنید.

۲. title فایل تمرین به صورت (نام تمرین+نام و نام خانوادگی) به انگلیسی باشد.

نمونه: AmirMohammadMohammadi_ansible_04

۳. در صورتی که سوال و یا ابهامی دارید در گروه چت تلگرامی بپرسید.

تمرین Ansible

مجموع امتیاز: چهل نمره

• تمام playbook ها روی سیستم Ubuntu 22.04 یا WSL/VM لوکال قابل اجرا هستند

• تحویل یک پوشه کامل شامل playbooks ، roles ، inventory و خروجی اجراها

پیش نیاز: تمرین Ansible بخش اول باید کامل شده باشد

ansible_playbook_homework/

└─ 01_playbook_basics/

└─ 02_playbook_advanced/

└─ 03_playbook_complex/

└─ 04_ansible_roles/

بخش اول — 01-ansible-playbook-basics - Playbook Basics: دوازده نمره

سوال ۱.۱ — اولین Playbook و Variables پنج نمره

آنچه می‌خواهم: نوشتن playbook و کار با variables و facts

فایل 01_playbook_basics/concepts.md با توضیح:

- **Playbook:** فایل YAML برای automation پیچیده، قابل استفاده مجدد و مستندسازی شده

- ساختار اصلی: name, hosts, become, vars, tasks

- **Variables:** playbook vars, inventory vars, command line -e, facts

- **Jinja2 syntax:** {{ variable_name }} برای استفاده از variables

- name: First Playbook with Variables

hosts: localhost

connection: local

gather_facts: yes

vars:

app_name: "MyApp"

app_version: "1.0.0"

app_port: 8080

app_home: "/opt/{{ app_name }}"

tasks:

- name: Display system info

debug:

msg: "Running {{ app_name }} v{{ app_version }} on
{{ ansible_hostname }}"

- name: Show system resources

debug:

msg: "System: {{ ansible_distribution }} | CPU:
{{ ansible_processor_vcpus }} | RAM: {{ ansible_memtotal_mb }}MB"

- name: Create app directory

file:

path: "{{ app_home }}"

state: directory

mode: '0755'

become: yes

ignore_errors: yes

- name: Deploy config file

copy:

content: |

APP_NAME={{ app_name }}

```
APP_VERSION={{ app_version }}
```

```
APP_PORT={{ app_port }}
```

```
HOSTNAME={{ ansible_hostname }}
```

```
DEPLOYED_AT={{ ansible_date_time.iso8601 }}
```

```
dest: "{{ app_home }}/config.env"
```

```
become: yes
```

```
ignore_errors: yes
```

```
- name: Set deployment facts
```

```
set_fact:
```

```
deployment_time: "{{ ansible_date_time.iso8601 }}"
```

```
deployment_user: "{{ ansible_user_id }}"
```

```
- name: Show deployment info
```

```
debug:
```

```
msg: "Deployed by {{ deployment_user }} at {{ deployment_time }}"
```

playbook: اجرای

ansible-playbook first_playbook.yml

```
ansible-playbook first_playbook.yml --syntax-check
```

```
ansible-playbook first_playbook.yml -e "app_port=9090 app_version=2.0.0"
```

سوال ۱.۲ Conditionals — و — Loops هفت نمره

آنچه می‌خواهیم: کار با شرط‌ها و حلقه‌ها

فایل: 01_playbook_basics/conditionals_loops.yml: ---

- name: Conditionals and Loops

hosts: localhost

connection: local

gather_facts: yes

vars:

packages:

- curl

- wget

- git

users:

- name: user1

state: present

- name: user2

state: present

tasks:

- name: Install packages on Debian

apt:

name: "{{ item }}"

state: present

update_cache: yes

loop: "{{ packages }}"

when: ansible_os_family == "Debian"

become: yes

ignore_errors: yes

- name: Create users

user:

name: "{{ item.name }}"

state: "{{ item.state }}"

shell: /bin/bash

loop: "{{ users }}"

become: yes

ignore_errors: yes

- name: Check directory

stat:

path: /opt/myapp

register: dir_check

- name: Create if not exists

file:

path: /opt/myapp

state: directory

when: not dir_check.stat.exists

become: yes

ignore_errors: yes

- name: Memory check

debug:

msg: "{{ 'Sufficient memory' if ansible_memtotal_mb > 1024 else 'Low memory' }}"

- name: Multiple conditions

debug:

msg: "Ubuntu with enough RAM"

when:

- ansible_distribution == "Ubuntu"

- ansible_memtotal_mb > 1024

فایل 01_playbook_basics/loops_explanation.md:

• when برای conditionals با <, >, !=, == operators:

• loop برای حلقه روی list، {{ item }} برای دسترسی

• register برای ذخیره نتیجه tasks

سوال ۲.۱ Handlers — و — Templates هشت نمره

آنچه می‌خواهم: استفاده از handlers و Jinja2 templates

فایل: 02_playbook_advanced/concepts.md

Handlers: tasks که فقط با notify و در صورت تغییر اجرا می‌شوند، مناسب برای restart services

Templates: فایل Jinja2 با: syntax

Variables: {{ var }}

Conditionals: {% if %} ... {% endif %}

Loops: {% for item in list %} ... {% endfor %}

پوشه: templates

templates/nginx.conf.j2:

user www-data;

worker_processes {{ ansible_processor_vcpus }};

http {

{% if enable_gzip %}

```
gzip on;
```

```
{% endif %}
```

```
server {
```

```
    listen {{ nginx_port }};
```

```
    server_name {{ server_name }};
```

```
    root {{ nginx_root }};
```

```
    location / {
```

```
        try_files $uri $uri/ =404;
```

```
    }
```

```
}
```

```
}
```

```
templates/app.conf.j2:
```

```
[general]
```

```
app_name={{ app_name }}
```

```
environment={{ environment }}
```

```
[server]

port={{ app_port }}


[database]

{% if environment == 'production' %}

db_host={{ prod_db_host }}

{% else %}

db_host={{ dev_db_host }}

{% endif %}


[users]

{% for user in app_users %}

{{ user.name }}={{ user.role }}

{% endfor %}
```

02_playbook_advanced/handlers_templates.yml: فایل

```
- name: Handlers and Templates Demo

hosts: localhost
```

connection: local

gather_facts: yes

become: yes

vars:

app_name: "MyApp"

app_version: "2.0.0"

app_port: 8080

environment: "development"

nginx_port: 80

nginx_root: /var/www/html

server_name: localhost

enable_gzip: yes

dev_db_host: "localhost"

prod_db_host: "db.example.com"

app_users:

- name: admin

role: administrator

- name: user1

role: user

tasks:

- name: Install nginx

apt:

name: nginx

state: present

ignore_errors: yes

- name: Generate nginx config

template:

src: templates/nginx.conf.j2

dest: /etc/nginx/nginx.conf

backup: yes

notify: Restart nginx

ignore_errors: yes

- name: Create app config directory

file:

path: /etc/myapp

state: directory

ignore_errors: yes

- name: Generate app config

template:

src: templates/app.conf.j2

dest: /etc/myapp/app.conf

ignore_errors: yes

- name: Deploy index page

copy:

content: |

<html><body>

<h1>{{ app_name }} v{{ app_version }}</h1>

<p>Deployed at {{ ansible_date_time.iso8601 }}</p>

```
</body></html>
```

```
dest: "{{ nginx_root }}/index.html"
```

```
notify: Reload nginx
```

```
ignore_errors: yes
```

```
handlers:
```

```
- name: Restart nginx
```

```
service:
```

```
name: nginx
```

```
state: restarted
```

```
ignore_errors: yes
```

```
- name: Reload nginx
```

```
service:
```

```
name: nginx
```

```
state: reloaded
```

```
ignore_errors: yes
```

سوال ۲.۲ — Error Handling — چهار نمره

فایل: 02_playbook_advanced/error_handling.yml

- name: Error Handling Demo

hosts: localhost

connection: local

gather_facts: yes

tasks:

- name: Task that fails but ignored

command: /bin/false

ignore_errors: yes

- name: Custom fail condition

shell: echo "Processing data"

register: result

failed_when: "'Error' in result.stdout"

- name: No change reported

command: echo "status check"

changed_when: false

- name: Debug variable

debug:

msg: "Hostname: {{ ansible_hostname }}, RAM:
{{ ansible_memtotal_mb }}MB"

- name: Assert requirements

assert:

that:

- ansible_memtotal_mb >= 512

- ansible_distribution in ["Ubuntu", "Debian"]

fail_msg: "Requirements not met"

success_msg: "All checks passed"

ignore_errors: yes

- name: Block rescue example

block:

- name: Risky task

command: /bin/false

rescue:

- name: Handle error

debug:

msg: "Error handled gracefully"

always:

- name: Cleanup

debug:

msg: "Always executed"

بخش سوم — Playbook Complex - ansible-playbook-3: دوازده نمره

سوال ۳.۱ Multi-Play — و — Import/Include هشت نمره

آنچه می‌خواهم playbook: با چند play و استفاده از import/include

فایل inventory: 03_playbook_complex/inventory

```
[webservers]
```

```
web1 ansible_host=localhost ansible_connection=local
```

```
[dbservers]
```

```
db1 ansible_host=localhost ansible_connection=local
```

```
[all:children]
```

```
webservers
```

```
dbservers
```

پوشه: tasks

```
tasks/install.yml:
```

```
- name: Update cache
```

```
  apt:
```

```
    update_cache: yes
```

```
  when: ansible_os_family == "Debian"
```

```
  ignore_errors: yes
```

- name: Install packages

apt:

name: "{{ item }}"

state: present

loop: "{{ packages }}"

ignore_errors: yes

tasks/configure.yml:

- name: Create directories

file:

path: "{{ item }}"

state: directory

loop: "{{ directories }}"

ignore_errors: yes

- name: Deploy config

copy:

content: "{{ config_content }}"

```
dest: "{{ config_path }}"
```

```
ignore_errors: yes
```

03_playbook_complex/multi_play.yml: فایل

```
---
```

```
# Play 1: Common setup
```

```
- name: Setup all servers
```

```
hosts: all
```

```
gather_facts: yes
```

```
become: yes
```

```
vars:
```

```
packages:
```

```
- curl
```

```
- wget
```

```
- vim
```

```
directories:
```

```
- /opt/apps
```

```
- /opt/logs
```

tasks:

- name: Show host

debug:

msg: "Configuring {{ ansible_hostname }}"

- name: Import package installation

import_tasks: tasks/install.yml

- name: Include configuration

include_tasks: tasks/configure.yml

vars:

config_content: "HOSTNAME={{ ansible_hostname }}\n"

config_path: "/opt/apps/host.conf"

Play 2: Web servers

- name: Configure web servers

hosts: webservers

become: yes

tasks:

- name: Install nginx

apt:

name: nginx

state: present

ignore_errors: yes

- name: Deploy web page

copy:

content: "<h1>Web Server {{ ansible_hostname }}</h1>"

dest: /var/www/html/index.html

ignore_errors: yes

Play 3: Database servers

- name: Configure database

hosts: dbservers

become: yes

tasks:

- name: Create db directory

file:

path: /var/lib/mydb

state: directory

ignore_errors: yes

- name: Create db config

copy:

content: "[mysqld]\nport=3306\n"

dest: /etc/my.cnf

ignore_errors: yes

03_playbook_complex/concepts.md: فایل

- Multi-play: چند play در یک playbook برای orchestration

- static: Import در parse time

- Include: dynamic در execution time

سوال ۳.۲ — Tags — چهار نمره

آنچه می‌خواهم: استفاده از tags برای کنترل اجرا

فایل: 03_playbook_complex/tags_demo.yml

- name: Tags Demo

hosts: localhost

become: yes

tasks:

- name: Always runs

debug:

msg: "This always executes"

tags: always

- name: Install web

apt:

name: nginx

state: present

tags:

- install

- webserver

ignore_errors: yes

- name: Install db

apt:

name: mysql-server

state: present

tags:

- install

- database

ignore_errors: yes

- name: Configure web

copy:

content: "# nginx config"

dest: /etc/nginx/nginx.conf

tags:

- configure

- webserver

ignore_errors: yes

- name: Configure db

copy:

content: "# mysql config"

dest: /etc/mysql/my.cnf

tags:

- configure

- database

ignore_errors: yes

- name: Never runs by default

debug:

msg: "Only with --tags never"

tags: never

استفاده:

```
ansible-playbook tags_demo.yml --tags install
```

```
ansible-playbook tags_demo.yml --tags webserver
```

```
ansible-playbook tags_demo.yml --skip-tags configure
```

```
ansible-playbook tags_demo.yml --list-tags
```

بخش چهارم — Ansible Roles - ansible-roles: یازده نمره

سوال ۴.۱ — ایجاد و استفاده از Roles: یازده نمره

آنچه می‌خواهم: ساختار role و استفاده در playbook با dependencies

فایل: 04_ansible_roles/roles_concepts.md

Role: ساختار استاندارد برای سازماندهی و reusability

ساختار:

role_name/

└─ tasks/main.yml # الزامی

```
|— handlers/main.yml
|— templates/
|— files/
|— vars/main.yml
|— defaults/main.yml
└— meta/main.yml    # برای dependencies
```

ایجاد:

```
cd 04_ansible_roles
```

```
mkdir -p roles
```

```
ansible-galaxy init roles/common
```

```
ansible-galaxy init roles/webserver
```

```
ansible-galaxy init roles/monitoring
```

Role: common

```
roles/common/tasks/main.yml:
```

```
---
```

```
- name: Install base packages
```

```
  apt:
```

```
    name:
```

- curl

- wget

- vim

state: present

become: yes

ignore_errors: yes

- name: Create directories

file:

path: "{{ item }}"

state: directory

loop:

- /opt/apps

- /opt/logs

become: yes

ignore_errors: yes

Role: webserver

roles/webserver/defaults/main.yml:

```
---

webserver_port: 80

webserver_root: /var/www/html

roles/webserver/tasks/main.yml:

---

- name: Install nginx

  apt:

    name: nginx

    state: present

  become: yes

  ignore_errors: yes


- name: Deploy index

  template:

    src: index.html.j2

    dest: "{{ webserver_root }}/index.html"

  become: yes

  notify: Restart nginx
```

ignore_errors: yes

- name: Start nginx

service:

name: nginx

state: started

become: yes

ignore_errors: yes

roles/webserver/handlers/main.yml:

- name: Restart nginx

service:

name: nginx

state: restarted

become: yes

ignore_errors: yes

roles/webserver/templates/index.html.j2:

<!DOCTYPE html>

```
<html>

<head><title>{{ ansible_hostname }}</title></head>

<body>

  <h1>{{ ansible_hostname }}</h1>

  <p>Port: {{ webserver_port }}</p>

  <p>Deployed: {{ ansible_date_time.iso8601 }}</p>

</body>

</html>
```

Role: monitoring

roles/monitoring/meta/main.yml:

dependencies:

- role: common

roles/monitoring/tasks/main.yml:

- name: Install monitoring tools

apt:

name:

- htop

- iotop

state: present

become: yes

ignore_errors: yes

- name: Create monitor script

copy:

content: |

#!/bin/bash

echo "=== System Status ==="

free -h

df -h /

dest: /opt/scripts/monitor.sh

mode: '0755'

become: yes

ignore_errors: yes

Roles: استفاده از

04_ansible_roles/deploy.yml:

- name: Deploy Application

hosts: localhost

connection: local

gather_facts: yes

vars:

environment: production

app_version: 1.0.0

pre_tasks:

- name: Show deployment info

debug:

msg: "Deploying v{{ app_version }} to {{ environment }}"

- name: Check requirements

assert:

that:

- ansible_memtotal_mb >= 1024

fail_msg: "Insufficient memory"

roles:

- role: common

tags: common

- role: webserver

vars:

webserver_port: 8080

tags: webserver

- role: monitoring

tags: monitoring

post_tasks:

- name: Verify deployment

uri:

url: http://localhost:8080

status_code: 200

register: health

failed_when: false

ignore_errors: yes

- name: Log deployment

copy:

content: |

Deployment: {{ app_version }}

Time: {{ ansible_date_time.iso8601 }}

Status: Success

dest: /opt/logs/deploy.log

become: yes

ignore_errors: yes

استفاده پیشرفته:

- name: Conditional Role Usage

hosts: localhost

tasks:

- name: Include webserver role

include_role:

name: webserver

when: ansible_distribution == "Ubuntu"

- name: Use role with custom vars

include_role:

name: monitoring

vars:

monitoring_interval: 60

تحويل نهایی — دو نمره

فایل: README.md

Ansible Playbooks و Roles

ساختار پروژه

- 01_*playbook*_basics: variables, conditionals, loops
- 02_*playbook*_advanced: handlers, templates, error handling
- 03_*playbook*_complex: multi-play, import/include, tags
- 04_*ansible*_roles: roles با dependencies

اجرا

ansible-playbook <playbook_name>.yaml

ansible-playbook <playbook>.yaml --tags <tag_name>

ansible-playbook <playbook>.yaml -e "var=value"

مفاهیم کلیدی

- Playbook structure: name, hosts, tasks
- Variables: vars, facts, set_*fact*
- Conditionals: *when*, *and*, *or*

item - *Loops: loop*

restart - *Handlers: notify*

{{ }}, *{% %}* - *Templates: Jinja2*

modular - *Roles: ساختار*

tasks - *Tags: کنترل اجرای*

فایل‌های خروجی:

- *find* یا *tree* *final_structure.txt* خروجی
- *playbook_runs.txt* تمام دستورات اجرا شده

معیار نمره‌دهی

- *20%* *syntax*: *Playbook structure*
- *15%* *loops*: *conditionals*, *Variables*
- *15%* *templates*: *Handlers*
- *15%* *import/include*: *Multi-play*
- *20%* *dependencies*: *Roles*

- ساختار فایل‌ها: ۱۰٪

- مستندسازی: ۵٪

مهلت تحویل: جمعه هفته آینده